

Raymarine Instrument User Survey

Thank you for participating in this survey. We are currently writing the specifications for the new ST70 Instrument Head and Autopilot Head.

These questions relate to some of the designs we are currently proposing. Your answers will help us to create a better product that's easier to use.

INSTRUCTIONS:

- This is **not** a test of you or your knowledge. The survey aims to find out how easy the interface designs are to understand. If you don't understand something please say so.
- Please give considered answers but there is no need to deliberate too long. Try to respond as you would if you had the device in front of you.
- Please answer the questions in the order they appear. Later questions contain information which gives away the answer for earlier questions. So if they are completed out of the given order, we won't get an accurate measure of the usability of the interface.

There are two parts to the test, one for each of the ST70 devices: Autopilot and Instruments.

Your responses will certainly help us to improve our product range. We look forward to hearing your thoughts.

Thank you,
Raymarine.

PART ONE: AUTOPILOT HEAD

QUESTION 1

At the start of calibration, the autopilot would ask you to select from the following options:

- Outboard Speed Boat
- Inboard Speed Boat
- Fly Bridge Cruiser
- Sports Cruiser
- Offshore sailboat
- Inshore sailboat
- Workboats (e.g. tugs, fishing vessels)

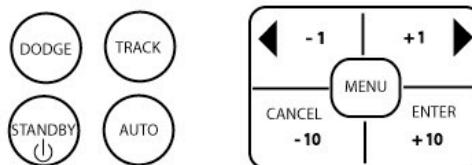
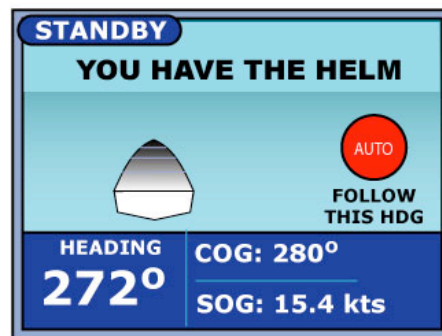
a) Do you understand what all of the items on this new list are? If not, which ones don't you understand?

b) Are there any other vessel types we should include on this list?

QUESTION 2.

For the next few questions, imagine you have just bought a new autopilot and ST70 Autopilot Head. You have performed the initial calibration. You are currently out in open water and you have previously plotted a course with several waypoints into your chart-plotter. You switch on the ST70 and you see the following screen:

PLEASE NOTE: You cannot press anything on the screen! You operate the ST70 via the buttons on the hardware below the screen. These buttons are shown below each screenshot.



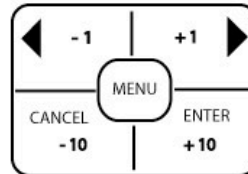
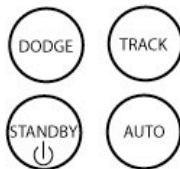
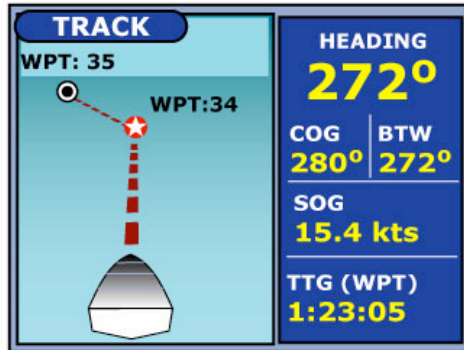
- a) Briefly describe your first impressions of the ST70 from this screen.

- b) Is the information presented clearly? Is it easy to see? Would you make any changes?

- c) Imagine that you want to activate the autopilot and get it to start guiding your vessel to the first waypoint you have entered into your chart-plotter. Which button do you press?

QUESTION 3

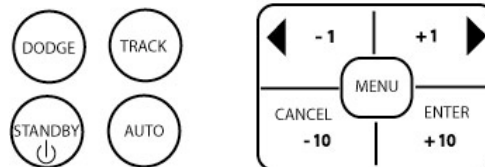
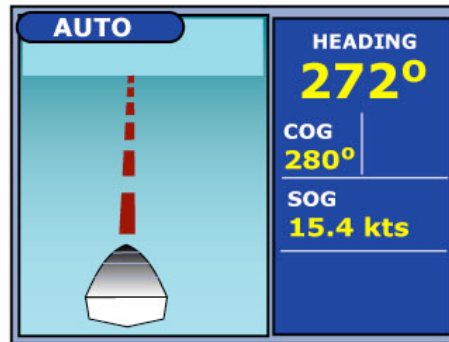
You hit the TRACK button and this activates the autopilot to start heading towards your first waypoint. The screen looks like this:



- a) Is the information on the screen clear enough? Would it be easy to see while onboard?
- b) What do you think of the colours used?
- d) What does TTG stand for?
- e) What do you think of the information presented on the screen? Is it useful? Is there other information you would rather see on this screen?
- f) You see an obstacle in the way on your current heading. It is quite close, you need to act quickly. What button would you press?
- g) What would you expect to happen when you pressed that button?
- e) You change your mind and decide not to follow your plotted course anymore. Instead you start heading towards your favourite fishing spot. You want the autopilot to keep this heading. Which button do you push?

QUESTION 4.

You have pushed AUTO and the boat is now heading along one direction. The screen appears like this:



- a) What do you think of the information presented? Is it useful? Can you think of other information you would rather see on this screen?

- b) What do you think of the graphic section on the left side of the screen?

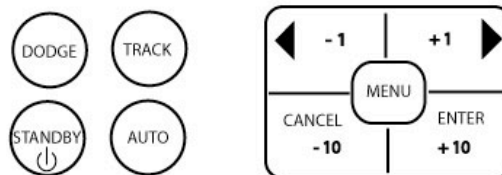
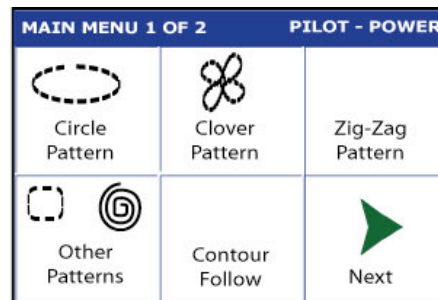
- c) The boat is staying on a heading of 272 degrees. If you wanted to change this by 15 degrees to port, what would you press?

- d) You are currently viewing the speed in knots. If you want to change this to OTHER OPTION?, how would you expect to do this?

- c) What would you expect to see if you pressed MENU? Give some examples of what you would expect to see listed.

QUESTION 6.

Let's suppose that you are on a powerboat. You are in AUTO mode, you pressed MENU and the screen now appears like this:



a) Briefly describe what information or functions you would expect from each of these options:

i) Circle Pattern

ii) Clover Pattern

iii) Ziz-Zag Pattern

iv) Other Patterns

v) Contour Follow

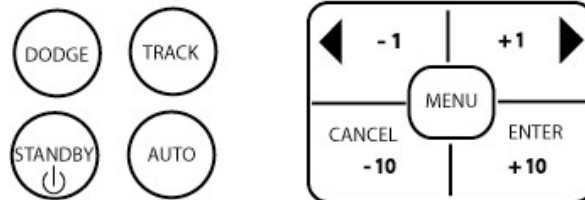
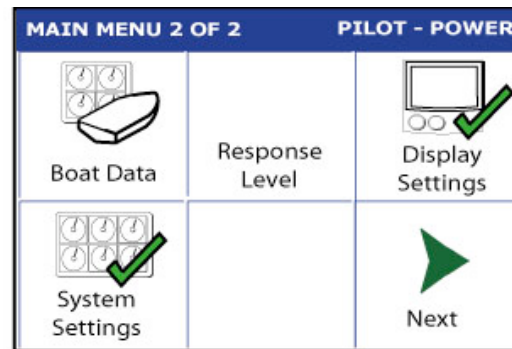
vi) Next

b) How well do you think the icon reflects the content for each item? (Choose from Poor, Good or Excellent).

- Circle Pattern
- Clover Pattern
- Ziz-Zag Pattern
- Other Patterns
- Contour Follow

QUESTION 7

You selected the 'Next' option on the previous screen, and the screen now appears like this:

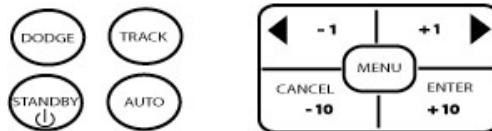
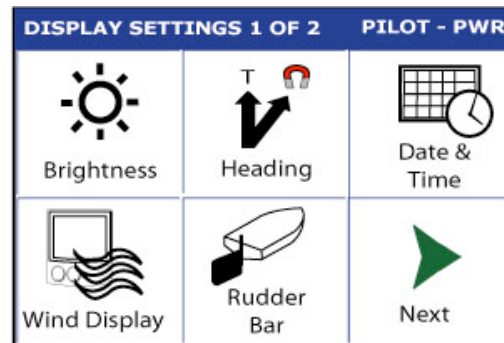


- a) Briefly describe what information or functions you would expect to find behind each item. Where the item includes settings, give some examples of settings you would expect.
- i) Boat Data
 - ii) Response Level
 - iii) Display Settings
 - iv) System Settings
- b) How well do you think the icons reflect the expected content of each option? (Choose from Poor, Good, Excellent)
- i. Boat Data
 - ii. Response Level
 - iii. Display Settings
 - iv. System Settings
- c) Looking at the last 2 screens presented, which option would you choose if you wanted to increase the Rudder Gain?

- d) When you first got the ST70 and autopilot you remember that you carried out a calibration process, by going through a calibration wizard. Looking at the last 2 screens presented, if you wanted to run the calibration wizard again, where would you expect to find it?

QUESTION 8

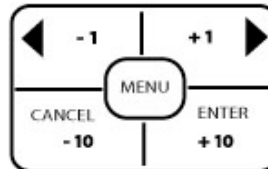
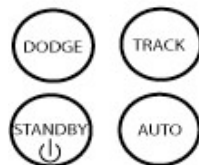
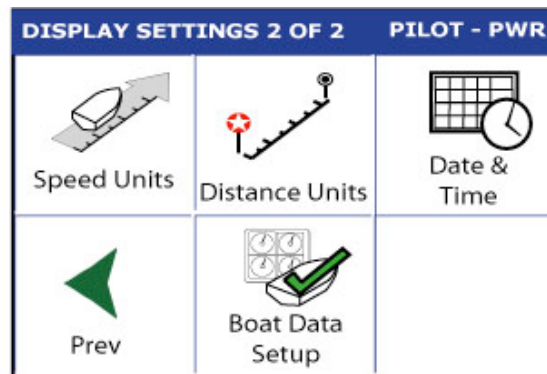
You have selected Display Settings from the previous screen, and the screen now looks like this:



- a) Briefly describe what you would expect to find behind each of these options:
- Brightness
 - Heading
 - Date & Time
 - Wind Display
 - Rudder Bar
- b) How well do you think each icon reflects the expected content:
- Brightness
 - Heading
 - Date & Time
 - Wind Display
 - Rudder Bar

QUESTION 9

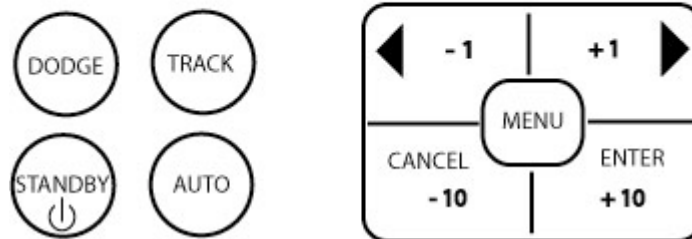
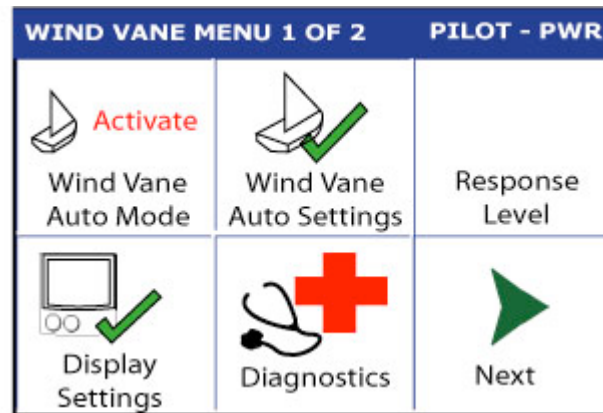
You selected the Next option on the previous screen, and the screen now looks like this:



- a) Briefly describe the information or functions you would expect to find behind each option:
- i) Unit Options
 - ii) Boat Data Setup
- b) How well do you think each icon reflects the expected content. (Choose from Poor, Good, Excellent):
- i) Unit Options
 - ii) Boat Data Setup

QUESTION 10

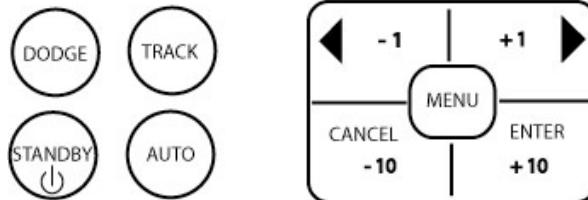
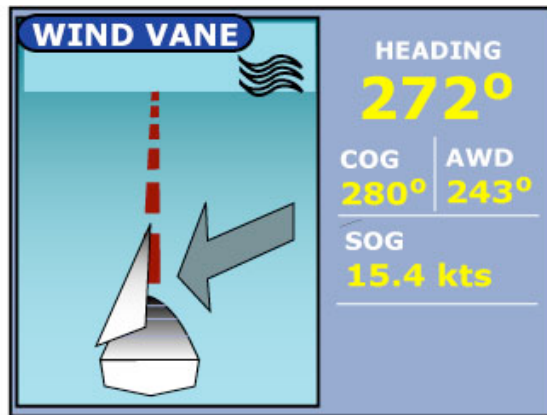
For the next few questions, imagine you have a sail boat. You want to put the autopilot into Wind Vane Mode, which keeps the boat on a particular angle to the wind. When you press MENU the screen looks like this:



- a) Briefly describe the information or functions you would expect to find behind the following options:
 - i) Wind Vane Auto Mode
 - ii) Wind Vane Auto Settings
 - iii) Diagnostics
- b) How well do you think each icon reflects the expected content?
 - i) Wind Vane Auto Mode
 - ii) Wind Vane Auto Settings
 - iii) Diagnostics

QUESTION 11

You activate the Wind Vane Mode, exit the menu and the screen now looks like this:

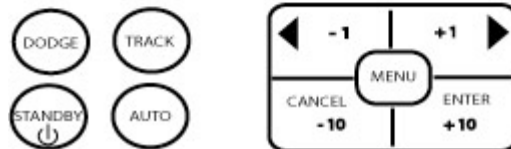


- What are your general impressions of this screen?
- What do you think of the information presented? Is there other information you would rather see?
- How would you carry out an Autotack?

QUESTION 12

You return to the Menu and then select the Wind Vane Auto Settings option. The screen now looks like this:

WIND VANE AUTO SETTINGS		SAIL
 Autotack Angle	 Autotack Delay	 Gybe Inhibit
Wind Trim		



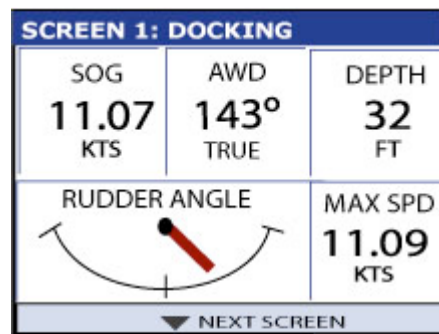
- a) Briefly describe what you information or function you would expect behind each of these options:
- Autotack Angle
 - Autotack Delay
 - Gybe Inhibit
 - Wind Trim
- b) How well does the icon reflect the content of each option?
- Autotack Angle
 - Autotack Delay
 - Gybe Inhibit
 - Wind Trim

PART TWO: ST70 INSTRUMENTATION

For this section of the survey, imagine you have purchased another ST70 device that you can use to display readings from the various transducers and instruments on your boat.

QUESTION 13

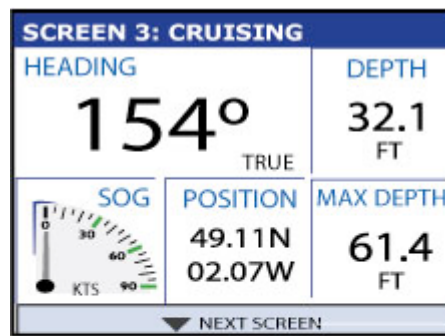
You have set up the ST70 Instrument Head and the screen now looks like this:



- What are your first impressions of the screen?
- Is the information displayed clearly? Would you make any changes?
- You know you can get up to 9 screens of instrument data displayed on this device. You are now looking at the first screen. How would you get to the next one?
- When you purchased the device, you were told that you could configure which instrument readouts appeared on each screen. Generally, how would you expect to access this functionality?

QUESTION 15

You press the down arrow to access other readout screens. The screen now looks like this:

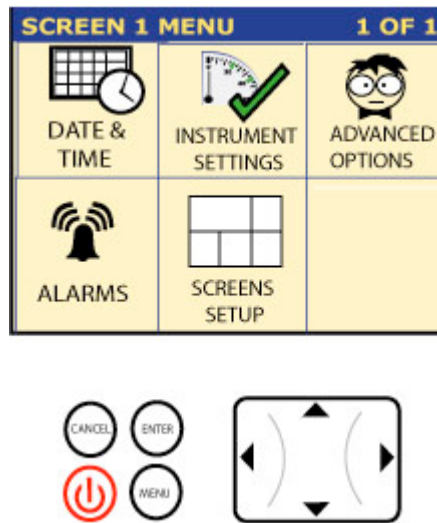


- a) Do you think the information here is presented clearly? Would you make any changes?

- b) If you wanted to reset the Max Depth reading, how would you expect to start going about this process?

QUESTION 15

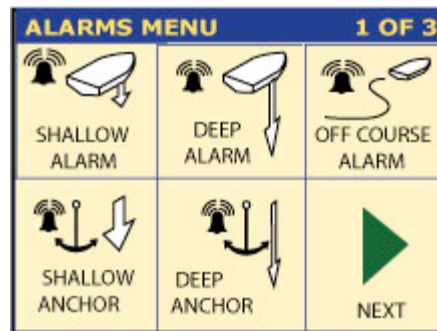
You press the MENU button and the screen now looks like this:



- a) Notice that the text in this menu is now in upper case whereas the Autopilot menus were in lower and upper case. Which do you prefer, upper case only or upper & lower together?
- b) Briefly describe what information and functions you would expect to find behind each of these options. Please give examples.
 - a. Instrument Settings
 - b. Advanced Options
 - c. Alarms
 - d. Screens Setup

QUESTION 16.

You select the ALARMS option from the previous menu. The screen now appears as follows:



- a) What are your impressions of this page? Are the icons easy to see?
- b) How well do you think the icons reflect the content of each option?:
 - a. Shallow Alarm
 - b. Deep Alarm
 - c. Off Course Alarm
 - d. Shallow Anchor Alarm
 - e. Deep Anchor Alarm
- c) When might you use the Deep Anchor Alarm?
- d) When might you use the Shallow Anchor Alarm?

QUESTION 17:

Without looking them up, do you know what each of these acronyms mean? Please write the meaning next to each one.

VMG

COG

SOG

BTW

CMG

DMG

AWA

TWA

TWD

TWS

XTE

SURVEY ENDS

Thank you for taking part in our survey. Your responses will help us to produce a better autopilot and instrumentation product.